UNIT 2

Machines & Measuring Instruments

Topics: DC generator • DC motor • Single-phase transformer • Three-phase induction motor • Alternator • Applications of machines • PMMC instrument • Moving-iron instrument • Wheatstone bridge

START HERE

Big picture: An **electrical machine** converts energy. A **generator** turns mechanical energy into electrical; a **motor** does the reverse; a **transformer** changes voltage level with no moving parts. **Measuring instruments** let us read current, voltage and resistance.

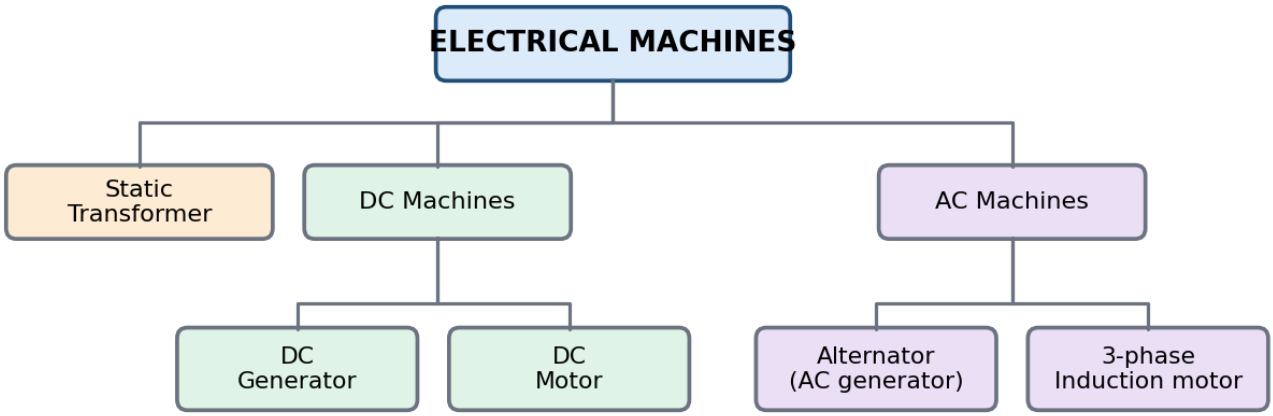


Fig 1 Classification of the machines in this unit

1 ELECTRICAL MACHINES

BASIC PRINCIPLES

Two golden principles

① **Faraday's law (generator action):** a conductor moving in a magnetic field (or a changing flux) induces an emf: e = B l v (or e = −N dΦ/dt).

② **Force on a current-carrying conductor (motor action):** F = B I l newton.

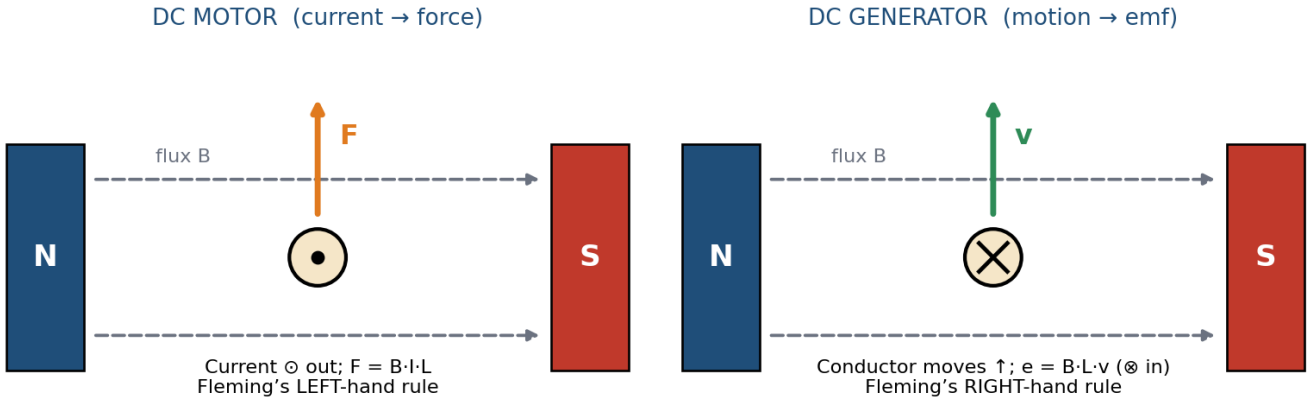


Fig 2 Motor action and generator action (same machine can do both)

1.1 DC Generator

Principle: converts mechanical energy into electrical energy (DC) by electromagnetic induction. The armature conductors cut the flux of the field poles and an emf is induced.

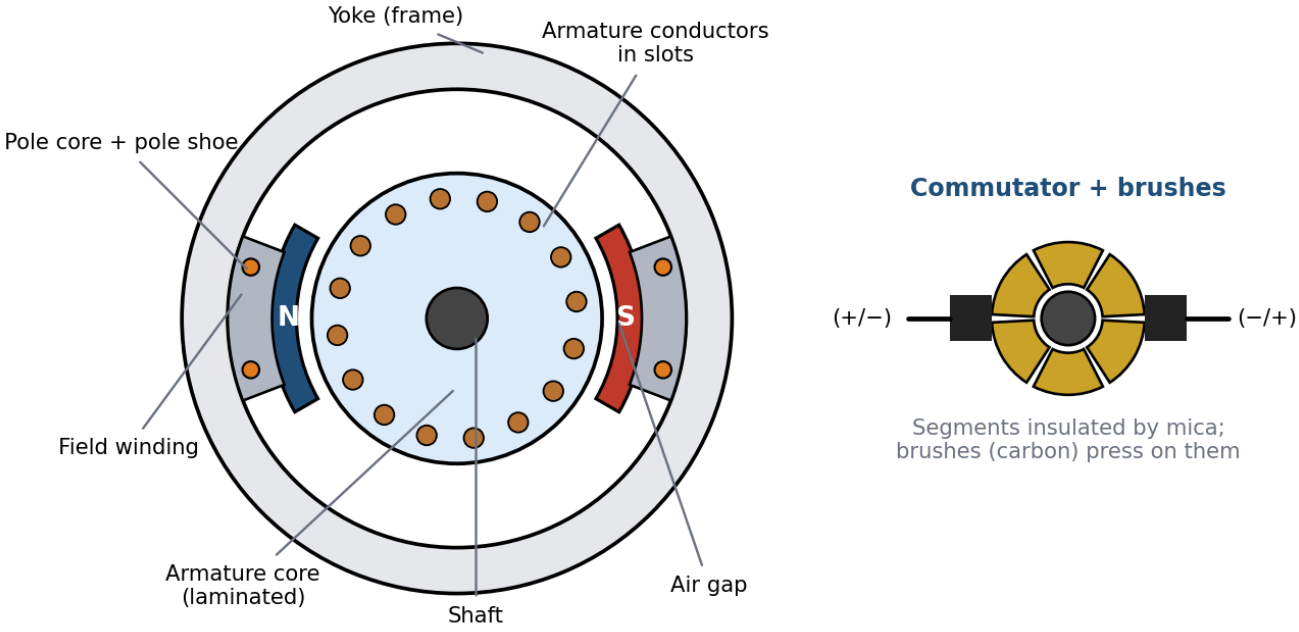


Fig 3 Cross-section of a DC machine and its commutator (same construction for generator and motor)

Part	Function
Yoke (frame)	Outer cast-iron/steel body; mechanical support and path for magnetic flux
Pole core + pole shoe	Carries field winding; shoe spreads flux evenly over the armature
Field winding	Copper coils on poles; produce the main magnetic field
Armature core + winding	Laminated steel core with slots; conductors in which emf is induced
Commutator	Copper segments; converts the alternating emf in the armature into **unidirectional** (DC) output
Brushes	Carbon blocks that collect current from the rotating commutator

$$E = (P \Phi Z N) / (60 A) \text{ volts}$$

P = number of poles, Φ = flux per pole (Wb), Z = total armature conductors, N = speed (rpm), A = number of parallel paths (A = P for lap winding, A = 2 for wave winding).

Type	Field connection	Typical applications
Separately excited	Field from an external DC source	Precise voltage control, test labs
Shunt	Field in **parallel** with armature	Battery charging, lighting, general supply
Series	Field in **series** with armature	Arc lamps, boosters
Compound	Both shunt and series fields	Lighting + power supply, welding sets

1.2 DC Motor

DEFINITION

Principle: when a current-carrying armature conductor is placed in a magnetic field it experiences a force; the force on the two sides of a coil forms a **torque** that rotates the armature. (Direction by Fleming's **left-hand rule**.)

$$T \propto \Phi I_a \quad E_b = V - I_a R_a \quad N \propto E_b / \Phi$$

Back emf (E_b): the rotating armature also generates an emf that *opposes* the supply voltage. It limits the armature current and makes the motor self-regulating — at start $E_b = 0$ so the current is very high, hence a **starter** is needed for large motors.

Motor	Characteristic	Applications
Shunt	Nearly **constant speed**	Lathes, fans, blowers, pumps, machine tools
Series	**High starting torque**; speed falls with load; never start on no-load (runaway)	Electric trains, cranes, hoists, elevators
Compound	Good starting torque + fairly steady speed	Rolling mills, presses, shears, punches

1.3 Single-Phase Transformer

DEFINITION
A **transformer** is a **static** device that transfers electrical energy from one circuit to another through **mutual induction**, changing voltage and current levels but keeping the **frequency** the same. It does **not** work on DC.

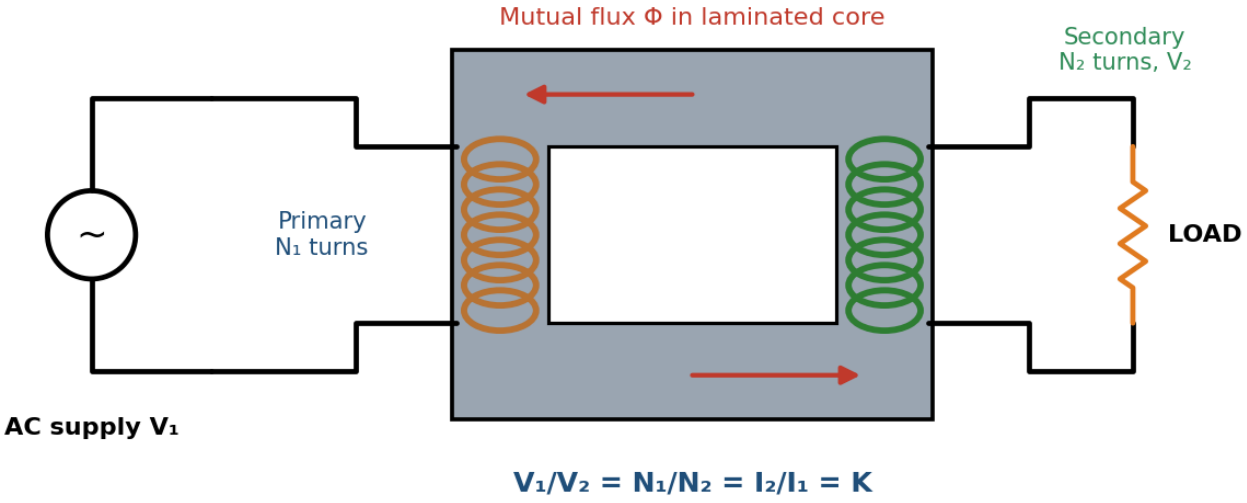


Fig 4 Single-phase transformer

Construction: a core of thin **laminated silicon-steel sheets** (reduces eddy-current loss) with two windings: **primary** (connected to supply) and **secondary** (connected to load). Types: *core type* and *shell type*. Small units are air-cooled; large ones are **oil-immersed**.
Working: AC in the primary produces an alternating flux Φ in the core → this flux links the secondary → emf is induced in the secondary by mutual induction.

$$E = 4.44\, f\, N\, \Phi_m \qquad \frac{V_1}{V_2} = \frac{N_1}{N_2} = \frac{I_2}{I_1} = K$$

Efficiency η = (Output power / Input power) × 100

Step-up transformer	Step-down transformer
$N_2 > N_1$, $K > 1$, $V_2 > V_1$, $I_2 < I_1$	$N_2 < N_1$, $K < 1$, $V_2 < V_1$, $I_2 > I_1$
Used at power stations (11 kV → 400 kV)	Used in distribution (11 kV → 415/230 V), adapters

REMEMBER
Losses: *Iron loss* (hysteresis + eddy current, constant) and *copper loss* (I^2R, varies with load). Ideal transformer: $V_1I_1 = V_2I_2$.

SOLVED EXAMPLE
Q1. A 2200/220 V transformer has 1000 primary turns. Find secondary turns and the primary current when it supplies 10 A.
$N_2 = N_1 \times V_2/V_1 = 1000 \times 220/2200 =$ **100 turns**. $I_1 = I_2 \times (V_2/V_1) = 10 \times 0.1 =$ **1 A**.

APPLICATIONS
Power transmission & distribution, mobile chargers, isolation, impedance matching, welding, voltage stabilisers.

1.4 Three-Phase Induction Motor

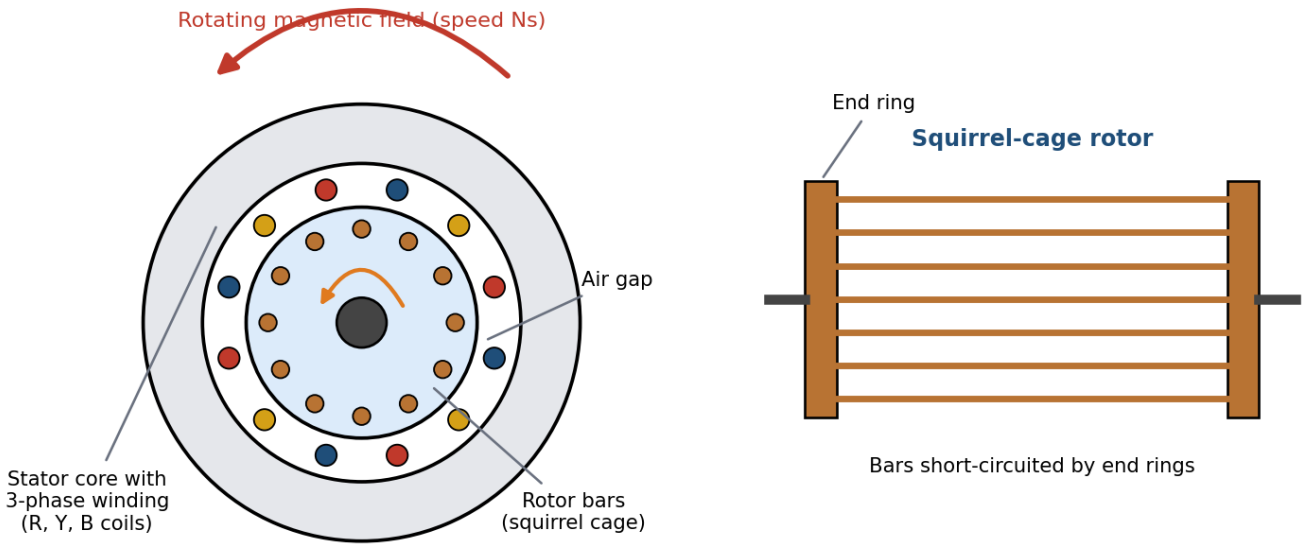


Fig 5 Stator and squirrel-cage rotor of a three-phase induction motor

DEFINITION

Principle: A three-phase supply to the stator winding creates a **rotating magnetic field** of synchronous speed N_s. It cuts the rotor conductors and induces emf → current → force (Lenz's law) → the rotor turns **in the direction of the field**, but always slightly slower. No electrical connection to the rotor, so it is also called an **asynchronous** motor.

$$N_s = 120\,f / P \text{ (rpm)} \quad \text{Slip } s = (N_s - N)/N_s$$

	Squirrel-cage rotor	Slip-ring (wound) rotor
Rotor	Copper/aluminium bars shorted by end rings	3-phase winding brought out to slip rings
Starting torque	Low	High (external resistance added)
Cost / maintenance	Cheap, rugged, no brushes	Costlier, needs brush maintenance
Uses	Pumps, fans, compressors, conveyors (≈90% of motors)	Cranes, lifts, crushers

SOLVED EXAMPLE

Q2. A 4-pole, 50 Hz induction motor runs at 1440 rpm. Find slip.

$N_s = 120 \times 50/4 = 1500$ rpm; $s = (1500 - 1440)/1500 = 0.04 =$ **4 %**.

1.5 Alternator (Synchronous Generator)

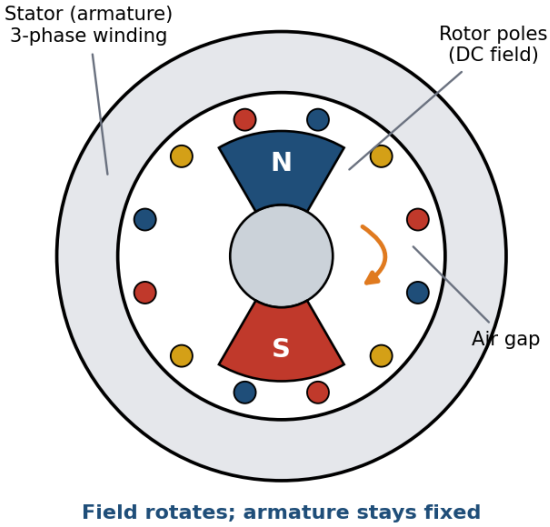


Fig 6a Cross-section of an alternator

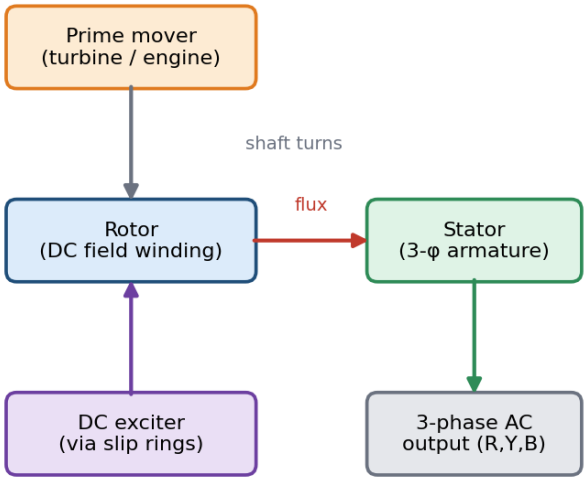


Fig 6b Block diagram of working

DEFINITION

Principle: Electromagnetic induction. The **rotor** carries a DC-excited field winding (supplied through slip rings); when turned by a prime mover its flux cuts the **stationary armature (stator)** conductors and induces a 3-phase AC emf.

$$f = P\,N / 120\ (Hz)\quad E_{ph} = 4.44\,K\,f\,\Phi\,T_{ph}$$

- **Why stationary armature?** Heavy high-voltage output (11 kV+) is easier to insulate and tap without slip rings; slip rings carry only the small DC field current.
- **Rotor types:** *Salient-pole* (low speed, hydro plants, many poles) and *cylindrical/non-salient* (high speed, steam/gas turbines, 2–4 poles).
- **Synchronous speed:** the rotor must run at N = 120f/P to deliver the correct frequency (50 Hz).

1.6 APPLICATIONS OF ELECTRICAL MACHINES

Machine	Typical applications
DC generator	Battery charging, electroplating, excitation of alternators, welding
DC motor	Electric traction (trains, trams, EVs), cranes, elevators, rolling mills, printing presses
Transformer	Power grid, adapters, inverters, isolation, instrument transformers
3-phase induction motor	Industrial drives, water pumps, fans, compressors, lifts, conveyors, washing machines
Alternator	Thermal, hydro and nuclear power stations, diesel generator sets, wind turbines, vehicle charging (alternator)

2 MEASURING INSTRUMENTS

Instruments are classified as **indicating** (show value on a scale), **recording** and **integrating** (energy meters). Every deflection instrument needs three torques: **deflecting**, **controlling** (spring) and **damping** (to stop oscillation).

2.1 Permanent Magnet Moving Coil (PMMC) Instrument

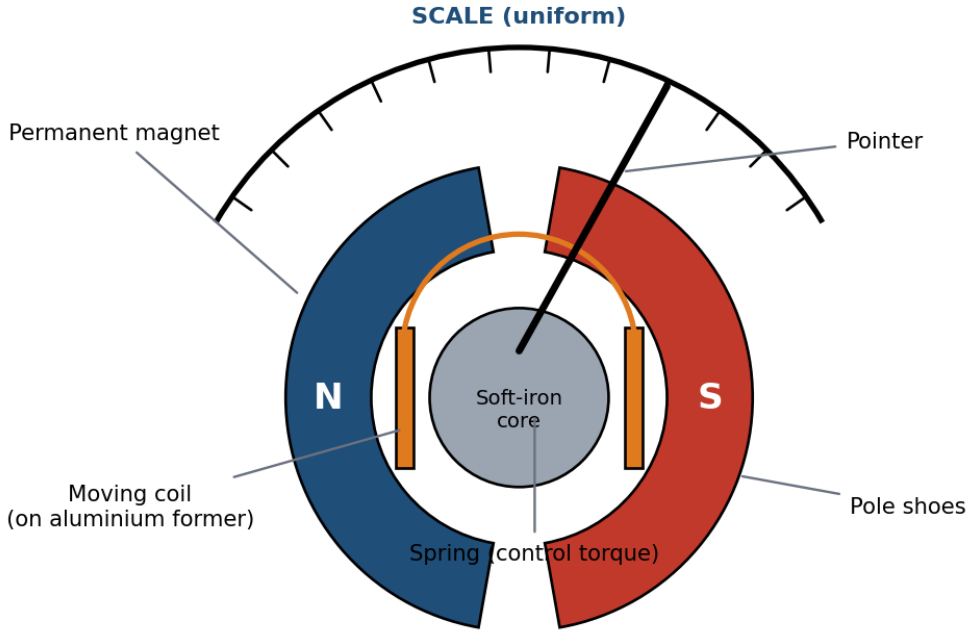


Fig 7 PMMC instrument

DEFINITION
Principle: a current-carrying coil placed in the magnetic field of a permanent magnet experiences a torque ∝ current. Deflection θ ∝ I, so the scale is **uniform**. Controlling torque: spiral **springs**. Damping: **eddy currents** in the aluminium former.

- **Advantages:** very accurate, uniform scale, high sensitivity, low power consumption, not affected by stray fields.
- **Disadvantages:** works on **DC only** (on AC the torque reverses and the pointer just vibrates), costlier, delicate.
- **Use:** DC ammeter (with parallel **shunt**) and DC voltmeter (with series **multiplier** resistor).

2.2 Moving Iron (MI) Instrument

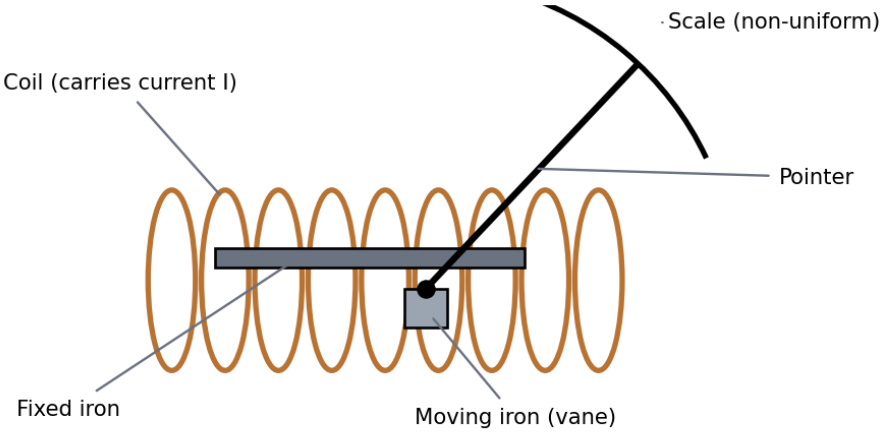


Fig 8 Moving-iron (attraction type) instrument

DEFINITION
Principle: a soft-iron piece placed in the field of a current-carrying coil gets magnetised and is **attracted** (attraction type) or two similar irons **repel** (repulsion type). Deflecting torque $\propto I^2$, so the scale is **non-uniform** (crowded at the start) but the instrument works on **both AC and DC**. Damping: air friction.

Feature	PMMC	Moving iron
Works on	DC only	AC and DC
Deflection $\propto$	I (uniform scale)	I^2 (non-uniform scale)
Accuracy / cost	High / costly	Moderate / cheap, rugged
Power consumption	Very low	Higher
Damping	Eddy current	Air friction

2.3 Wheatstone Bridge

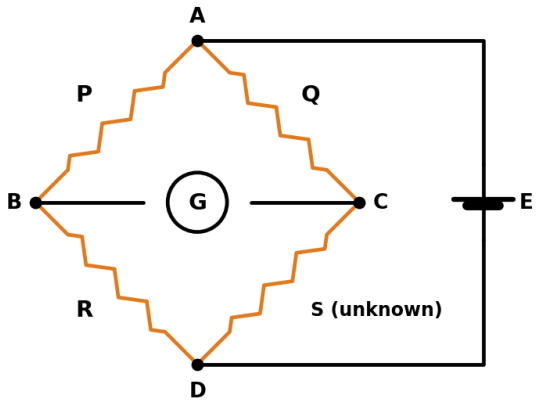


Fig 9 Wheatstone bridge

Use: measuring an unknown medium resistance accurately by comparison with known resistances.

Balance: adjust R until the galvanometer shows **zero** deflection. Then $V_B = V_C$ and the ratio arms satisfy the condition below.

P, Q = ratio arms; R = variable known resistance; S = unknown resistance.

$$P / Q = R / S \Rightarrow S = R \times Q / P$$

SOLVED EXAMPLE
Q3. At balance P = 100 Ω, Q = 1000 Ω, R = 35 Ω. Then S = 35 × 1000/100 = **350 Ω**.

- **Applications:** precise resistance measurement, strain gauges, temperature (RTD) and sensor circuits.
- **Limitations:** unsuitable for very low or very high resistance; heating by current may alter resistance.

3 QUICK REVISION & EXAM CORNER

Topic	Remember
DC generator emf	E = PΦZN/60A; commutator gives DC
DC motor	Eb = V − IaRa; T ∝ ΦIa; series = high starting torque
Transformer	$V_1/V_2 = N_1/N_2 = I_2/I_1$; E = 4.44fNΦm; no DC, no frequency change
Induction motor	Ns = 120f/P; slip = (Ns − N)/Ns; rotor never reaches Ns
Alternator	f = PN/120; rotating field, stationary armature
PMMC vs MI	PMMC: DC, uniform scale; MI: AC/DC, I^2 scale
Wheatstone	P/Q = R/S at null deflection

Short questions (2 marks)

- State the working principle of a DC motor and name Fleming's rule used.
- What is the function of a commutator?
- What is back emf? Why is a starter needed for a DC motor?
- Why is the transformer core laminated?
- Define slip of an induction motor.
- Why do alternators have a stationary armature?
- Give two advantages of PMMC instruments.
- Write the balance condition of a Wheatstone bridge.

Long questions (5–10 marks)

- Explain construction and working of a DC generator with a neat diagram; derive its emf equation.
- Explain the construction and working principle of a single-phase transformer; derive E = 4.44 f N Φm.
- Explain the production of a rotating magnetic field and working of a three-phase induction motor.
- Describe the construction and working of an alternator.
- With neat diagrams compare PMMC and moving-iron instruments.
- Explain the Wheatstone bridge and derive its balance condition.

Practice numericals (with answers)

No.	Problem	Answer
1	A 6-pole DC generator runs at 1000 rpm, Φ = 0.02 Wb, Z = 300, lap winding. Find emf.	E = 6×0.02×300×1000/(60×6) = 100 V
2	A transformer has N_1 = 500, N_2 = 50, V_1 = 440 V. Find V_2.	44 V
3	A 6-pole, 50 Hz induction motor has slip 5 %. Find rotor speed.	N_s = 1000 rpm; N = 950 rpm
4	Alternator with 4 poles must give 50 Hz. Find speed.	1500 rpm
5	DC motor: V = 230 V, I_a = 20 A, R_a = 0.5 Ω. Find E_b.	220 V

Quick MCQs

Question	Options	Ans
1. A transformer works on	(a) DC (b) AC (c) both (d) neither	b
2. Induction motor runs at … synchronous speed	(a) above (b) equal to (c) below (d) double	c
3. Speed of a series motor on no-load is	(a) zero (b) constant (c) dangerously high (d) low	c
4. Scale of an MI instrument is	(a) uniform (b) non-uniform (c) logarithmic (d) none	b